Department of
Computer Science and Engineering

Title: C - Java Syntax Similarity : Array, Conditionals, Loops

Object Oriented Programming

CSE 202

Green University of Bangladesh

1 Objective(s)

- To gather knowledge of java syntax, array conditions and loops.
- To implement different types of problem like prime number, Fibonacci number, odd even number checking solved in Java.

2 Problem analysis

In structured programming we learn different types of problem like prime number generation, Fibonacci number generation, checking odd and even numbers. In this lab, we will solve all types of this problem in java programming language, .

3 Prime Number Generation Code Implementation in Java

```java
/* Prime number generation code in java */
import java.util.Scanner;
public class Rayhan
{
 public static void main(String args[])
 {
 int num, i, count=0;
 Scanner scan = new Scanner(System.in);
 System.out.print("Enter a Number = ");
 num = scan.nextInt();
 for(i=2; i<num; i++)
 {
 if(num%i == 0)
 {
 count++;
 break;
 }
 }
 if(count == 0)
 {
 System.out.println("This is a Prime Number");
 }
 else
 {
 System.out.println("This is not a Prime Number");
 }
 }
}
```

4 Input/Output

Output of the program is given below.

```
Enter a number: 5
This is a Prime Number
Enter a number: 10
This is not a Prime Number
```

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5 Fibonacci Number Generation Code Implementation in Java

```
/* Fibonacci number generation code in java */
import java.util.Scanner;
public class Rayhan
 Page | 3
{
 public static void main(String[] args)
 {
 int n, a = 0, b = 0, c = 1;
 Scanner s = new Scanner(System.in);
 System.out.print("Enter value of n = ");
 n = s.nextInt();
 System.out.print("Fibonacci Series = ");
 for(int i = 1; i <= n; i++)
 {
 a = b;
 b = c;
 c = a + b;
 System.out.print(a+" ");
 }
 }
}
```

6 Input/Output

Output of the program is given below.

```
Enter value of n: 5
Fibonacci Series= 0 1 1 2 3
Enter value of n: 9
Fibonacci Series= 0 1 1 2 3 5 8 13 21
```

7 Discussion & Conclusion

From this experiments we learn about how to write a program in java. The different types of problem which we were practicing in structured programming, we just try to solve this using java language.

8 Lab Task (Please implement yourself and show the output to the instructor)

1. Implement summation of factorial number in java.

8.1 Problem analysis

Enter a number from user. After taking the number find the factorial of the particular number. According to the summation equation, we have to find out the summuation of factorial numbers and should be printed on screen. **However, Input must be taken from users.**

9 Lab Exercise (Submit as a report)

- Implement checking of odd and even number

$$\text{Sum} = \frac{x^2}{1!} + \frac{x^4}{3!} + \frac{x^6}{5!} + \ldots \frac{x^n}{(n-1)!}$$

Figure 1: Problem equation

- Implement summation of factorial odd number series.

10 Policy

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